Name: Muhammad Hamza   Enrollment: 01-131232-057

Architecture Quality Checklist

Assignment 01 – Cloud Architecture Design

Before submitting your assignment, verify that your work satisfies the following quality checks. Also submit hardcopy of this checklist.

1. Problem Understanding

☒ The system purpose is clearly explained.
☒ The target users and system environment are identified.
☒ Functional requirements describe what the system must do.
☒ Non-functional requirements include scalability, security, and availability.
☒ Each requirement is connected to an architectural implication.

Quick self-check:
If someone reads only Section 1, can they clearly understand what system is being designed?

2. Delivery Model Design

☒ The system is broken into multiple components.
☒ Each component is assigned an appropriate delivery model (IaaS / PaaS / SaaS).
☒ At least **3-4 components** are analyzed.
☒ Each delivery model choice is technically justified.
☒ A shared responsibility matrix is included.

Quick self-check:
Did you avoid assigning a single delivery model to the entire system?

3. Deployment Model Decision

☒ A deployment model is clearly selected.
☒ Justification includes data sensitivity considerations.
☒ Governance and compliance factors are discussed.
☒ Cost implications are explained.
☒ Scalability requirements are considered.

Quick self-check:
Did you explain *why this model is appropriate*, not just what it is?

4. Architecture Diagram Quality

☒ Diagram clearly shows system layers.
☒ Major components are labeled.
☒ Delivery model for components is indicated.
☒ Deployment boundaries are shown.
☒ Trust boundaries are illustrated.
☒ Security components are included.

Quick self-check:
If someone sees only the diagram, can they understand how the system works?

5. Cloud Characteristics Implementation

☒ Each cloud characteristic is linked to a specific part of the architecture.
☐ Explanation focuses on implementation, not definition.
☐ Operational impact is discussed.
☒ Architecture supports scalability and reliability.

Quick self-check:
Did you explain *how the architecture enables the characteristic*?

6. Risk and Governance Analysis

☒ At least **five risks** are identified.
☒ Each risk has a root cause explanation.
☐ Each risk includes architectural impact.
☒ Each risk includes a mitigation strategy.
☒ Responsibility (provider vs consumer) is identified.

Quick self-check:
Are your risks specific to your architecture?

7. Engineering Reasoning

☒ Architectural decisions are logically justified.
☒ Trade-offs are acknowledged.
☒ Cost vs control considerations are discussed.
☒ Limitations of the design are recognized.

Quick self-check:
Did you explain **why your architecture makes sense**, not just what it is?

8. Academic Integrity

☒ All explanations are written in your own words.
☒ No diagrams are copied directly from the internet.
☒ External references (if used) are properly cited.

9. Formatting Compliance

☒ Report length is **7–8 pages maximum**.
☒ Sections follow the provided template.
☒ Font and spacing follow assignment instructions.
☒ Tables and diagrams are readable.

10. Final Engineering Check

Answer these questions in Yes/No/Not Clear:

- Can this architecture **scale if user traffic increases?**

  Yes – Auto-scaling groups and load balancers

- Can it **handle failures without major disruption?**

Yes – Multi-zone and multi-region deploymen

- Does it **protect sensitive data properly?**

  Yes – Payment/user data isolated in private cl

- Does it **balance cost and control effectively?**

  Yes – Public cloud reduces cost; private cloud

If the answer to these questions is **no/not clear**, improve your design.

Remarks: *(If there is anything different that you would like to explain about your assignment, explain here)*

~~Cloud characteristics section could be improved with deeper operational impact discussi~~